

Original Report

Influence of planning time and treatment complexity on radiation therapy errors



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Abstract

Purpose: Radiation treatment planning is a complex process with potential for error. We hypothesized that shorter time from simulation to treatment would result in rushed work and higher incidence of errors. We examined treatment planning factors predictive for near-miss events.

Methods and materials: Treatments delivered from March 2012 through October 2014 were analyzed. Near-miss events were prospectively recorded and coded for severity on a 0 to 4 scale; only grade 3-4 (potentially severe/critical) events were studied in this report. For 4 treatment types (3-dimensional conformal, intensity modulated radiation therapy, stereotactic body radiation therapy [SBRT], neutron), logistic regression was performed to test influence of treatment planning time and clinical variables on near-miss events.

Results: There were 2257 treatment courses during the study period, with 322 grade 3-4 near-miss events. SBRT treatments had more frequent events than the other 3 treatment types (18% vs 11%, $P = .04$). For the 3-dimensional conformal group (1354 treatments), univariate analysis showed several factors predictive of near-miss events: longer time from simulation to first treatment ($P = .01$), treatment of primary site versus metastasis ($P < .001$), longer treatment course ($P < .001$), and pediatric versus adult patient ($P = .002$). However, on multivariate regression only pediatric versus adult patient remained predictive of events ($P = 0.02$). For the intensity modulated radiation therapy, SBRT, and neutron groups, time between simulation and first treatment was not found to be predictive of near-miss events on univariate or multivariate regression.

Conclusions: When controlling for treatment technique and other clinical factors, there was no relationship between time spent in radiation treatment planning and near-miss events. SBRT and pediatric treatments were more error-prone, indicating that clinical and technical complexity of treatments should be taken into account when targeting safety interventions.

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Introduction

Radiation treatment planning is an essential step in radiation therapy. It is a complex process, with many interlocking steps and many staff members involved.

Conflicts of interest: None.

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Treatment planning starts with the simulation appointment and ends with the final checks on the treatment machine before the first fraction of treatment. Recent studies have demonstrated that this process is error-prone, with potentially catastrophic results for the patient.¹ Researchers have begun to study the causes of radiation planning errors and potential interventions, such as standardized procedures and checklists.^{2–5} Several risk factors for radiation therapy errors have been identified, including shorter time from plan approval to first treatment, number of prescription items, pediatric patient, and treatment complexity.^{4,5}

Because each step in treatment planning cannot proceed until the prior step is complete, overly optimistic choice of treatment start date or delays in the initial phases such as target volume contouring can have downstream effects and result in rushing in later steps. Anecdotally, we have noted that time pressure in planning is a source of stress for staff in many radiation oncology departments. This time pressure may also lead to increased errors. In other fields of medicine, there is evidence of this effect: In 2 studies of medication prescribing errors by trainee physicians, time pressures/rushing were found to be major contributors to errors.^{6,7} A recent study in an academic radiation oncology department found similar patterns, with increased rate of treatment planning incidents when planning time was 1 day or less.⁵ Some centers have developed techniques to reduce process time and limit variability using statistical process control.⁸

In early 2012, our department implemented an incident learning system. A nonpunitive environment has been created; there has been a high level of staff buy-in and high volume of reports, which indicates a robust safety culture.⁹ The goal of the present study was to use these reports to examine the influence of time spent in each part of the treatment planning process on errors, while controlling for confounding factors such as treatment technique and clinical situation. Our hypothesis was that shorter time from simulation to treatment would result in rushed work and higher incidence of severe near-miss events.

Methods and materials

Extraction of treatment information

We analyzed all computed tomography–planned external beam radiation treatments delivered in our department from March 2012 to October 2014. Total body irradiation and electron treatments were excluded because of not having a computed tomography–based treatment plan, making it difficult to accurately determine the dates of completion of treatment planning steps. Treatment information was extracted using a database query from the department’s oncology information system (OIS), MOSAIQ version 2.60 (Elekta, Stockholm, Sweden). The unit of analysis was a treatment course, defined as continuous treatment of 1 or more anatomic sites. Therefore, a breast cancer treatment with initial and boost phases would be grouped into 1 treatment course, as would simultaneous treatment of 2 different bone metastases. The dates of treatment planning steps were extracted using the information listed in Table 1.

A very small proportion (3%) of treatment courses were excluded from analysis because dates for all steps could not be determined or dates were out of order. Because in our workflow physics chart, check/quality assurance can be performed either before or after verification simulation, it fell outside of the linear process model used in this study and was not included in the main analysis. A separate analysis showed that time from plan completion to physics chart check, and from physics chart check to first treatment did not affect the likelihood of near-miss events (data not shown).

Incident reporting

The department incident learning system is used to prospectively track events occurring in any part of the treatment process. The database is designed to be broadly inclusive of events affecting patient care, so a large spectrum of event severity is included, ranging from minor issues such as scheduling delays to serious issues such as incorrect dose or treatment site selected before being caught at a later step. This practice is consistent with previous studies.¹⁰ Staff members are able to enter reports from any department computer. Each week, new reports are reviewed at an interdisciplinary meeting and graded for severity on a 0 to 4 scale as described by another publication from our group.¹¹ For this analysis, only grade 3–4 near-miss events were considered, with the rationale that these events are most directly related to patient safety. Near-miss events were linked to treatment courses by finding events for that patient that were reported between date of simulation and 7 days after the last treatment. Events reported all the way through treatment were included because errors that originate in treatment

Table 1 Extraction of dates of planning steps from OIS

Endpoint	Data field in OIS
Simulation	Simulation billing charge
Contour submission from physician to dosimetry	Quality checklist item submitted by physician, or plan assigned to position in dosimetry queue
Plan completed by dosimetry	Treatment plan uploaded to OIS
Verification simulation/isocenter marking	Verification simulation billing charge
First treatment	Dose delivery recorded in OIS

OIS, oncology information system.

Table 2 Patient and treatment characteristics

	N	%
Treatment type		
3D conformal	1354	60.0
IMRT	675	29.9
SBRT	109	4.8
Neutron	119	5.3
Treatment site		
Primary tumor	1572	69.7
Metastasis	685	30.3
	Median	Interquartile range
Age	59	48-67
Number of fractions	14	5-30
Days from simulation to first treatment	6	2-11
Days from simulation to contour submission	1	0-2
Days from contour submission to plan completion	2	1-4
Days from plan completion to verification simulation	1	1-2
Days from verification simulation to first treatment	0	0-1

3D, 3-dimensional; IMRT, intensity modulated radiation therapy; SBRT, stereotactic body radiation therapy.

planning are often not caught until the patient is on treatment.¹²

Analysis

The goal of the analysis was to examine the influence of clinical and technical factors, including time spent in treatment planning steps, on near-miss events. In terms of

workload, the process of planning a palliative 3-dimensional (3D) conformal treatment is very different from that for planning for a more complex treatment such as stereotactic body radiation therapy (SBRT). A planning time of 2 days would not be considered rushed for a simple palliative treatment, but would be for SBRT. Therefore, we performed separate analyses for 4 treatment techniques: 3D conformal, intensity modulated radiation therapy (IMRT), SBRT, and neutron. For each treatment type, univariate and multivariate logistic regression predicting the number of grade 3-4 near-miss events were performed. The unit of analysis was a treatment course. The dependent variable was presence/absence of grade 3-4 near-miss event. Predictor variables included time from simulation to first treatment as well as other factors that could influence errors: primary tumor versus metastasis treated, number of fractions, and pediatric versus adult treatment. For the 3D conformal and IMRT groups, there was a large enough sample size to perform a second analysis including the time spent in the individual planning steps (contouring, dosimetry planning, etc.).

Only weekdays were counted for the time gap calculations because little work is done on the weekends for most patients. For purposes of analysis, steps occurring on the weekend were assumed to have occurred on the previous Friday (this affected 7% of treatment courses). Treatment site was coded as primary tumor or metastasis as follows: International Classification of Diseases-9 codes 197 and 198 were coded as treatment of metastasis; all others were coded as primary tumor. For analysis of the entire patient population, the Wilcoxon rank-sum test and χ^2 test were used.

A *P* value significance cutoff of .05 was used. Statistics were computed in R, version 3.0.2 (University of Auckland).

Results

Treatment type

There were 2257 treatment courses in 1975 patients during the 32-month study period. The majority underwent 3D conformal therapy (60%) or IMRT (30%). Table 2 lists treatment and patient characteristics. There were 1842 near-miss events in 1035 treatment courses. This analysis focuses on 322 severe/grade 3-4 events, which made up 17% of all events. These 322 events occurred in 267 distinct treatment courses because some treatment courses had more than 1 event. There was a significant difference in number of events between treatment types: SBRT had the highest proportion of treatment courses with grade 3-4 events (18%), followed by neutron (13%), 3D conformal (12%), and IMRT (11%); *P* = .04 for SBRT versus others. There were more grade 3-4 events when the primary tumor was treated compared with a metastasis (13% vs 9%, *P* =

Table 3 Predictors of time from simulation to first treatment (multivariate linear regression)

Characteristic	Beta coefficient (95% CI)	<i>P</i> value
Treatment fractions (per fraction)	0.11 (0.09-0.13)	<.001
Pediatric patient	1.06 (0.24-1.88)	.01
Treatment type (relative to 3D conformal)		
IMRT	3.22 (2.70-3.75)	<.001
SBRT	9.55 (8.63-10.48)	<.001
Neutron	7.07 (6.19-7.95)	<.001
Metastasis treated	-1.48 (-1.98 to -0.98)	<.001

Positive beta indicates longer time from simulation to first treatment. 3D, 3-dimensional; CI, confidence interval; IMRT, intensity modulated radiation therapy; SBRT, stereotactic body radiation therapy.

Table 4 Predictors of treatments with grade 3-4 near-miss events

Treatment type	Characteristic	Univariate		Multivariate	
		OR (95% CI)	<i>P</i> value	OR (95% CI)	<i>P</i> value
3D conformal (n = 1354)	Days from simulation to first treatment	1.04 (1.01-1.06)	.01	1.01 (0.98-1.04)	.41
	Metastasis treated	0.52 (0.36-0.74)	<.001	0.71 (0.46-1.08)	.11
	Treatment fractions	1.03 (1.01-1.04)	<.001	1.02 (1.00-1.03)	.07
	Pediatric patient	2.25 (1.33-3.66)	.002	1.84 (1.07-3.04)	.02
IMRT (n = 675)	Days from simulation to first treatment	1.05 (0.99-1.11)	.07	1.05 (0.99-1.11)	.07
	Metastasis treated	1.19 (0.40-2.89)	.73	1.15 (0.36-3.13)	.80
	Treatment fractions	1.00 (0.98-1.02)	.98	1.00 (0.98-1.02)	.82
	Pediatric patient	0.54 (0.09-1.84)	.41	0.54 (0.09-1.90)	.42
SBRT (n = 109) ^a	Days from simulation to first treatment	1.04 (0.96-1.12)	.32	1.04 (0.96-1.12)	.34
	Metastasis treated	0.87 (0.30-2.35)	.79	0.95 (0.32-2.65)	.92
	Treatment fractions	1.05 (0.78-1.39)	.75	1.05 (0.78-1.40)	.75
Neutron (n = 119) ^a	Days from simulation to first treatment	0.95 (0.85-1.05)	.28	0.95 (0.85-1.05)	.32
	Metastasis treated	0.79 (0.12-3.20)	.77	0.24 (0.02-1.61)	.18
	Treatment fractions	0.90 (0.76-1.07)	.23	0.84 (0.67-1.06)	.14

Univariate and multivariate logistic regression results are shown. A separate regression was performed for each of the four treatment techniques.

Bolded *P* value indicates $P < .05$.

3D, 3-dimensional; CI, confidence interval; IMRT, intensity modulated radiation therapy; OR, odds ratio; SBRT, stereotactic body radiation therapy.

^a Pediatric patient variable was not included for neutron and SBRT treatments because of the small number of pediatric patients (n = 2) receiving SBRT causing variable separation; no pediatric patients received neutron treatment.

.009). There was a positive correlation between number of fractions and grade 3-4 events, but with a small effect size (median fractions = 16 vs 14 for treatments with/without grade 3-4 event; $P = .04$). Treatments in pediatric patients had more grade 3-4 events than adult treatments (18% vs 11%; $P = .04$).

Time in planning steps

On multivariate linear regression, time spent in treatment planning was found to be highly influenced by factors such as treatment technique (Table 3). This reinforced the decision to perform separate analyses of near-miss events in the 4 treatment technique groups. For each treatment technique (3D conformal, IMRT, SBRT, neutron), we performed univariate and multivariate logistic regression predicting whether a treatment course had a grade 3-4 near-miss event. Results are presented in Table 4. The primary measure of time pressure in treatment planning was time from simulation to first treatment. For the 3D conformal group (1354 treatments), univariate analysis showed several factors predictive of near-miss events: longer time from simulation to first treatment ($P = .01$), treatment of primary site versus metastasis ($P < .001$), longer treatment course ($P < .001$), and pediatric versus adult patient ($P = .002$). However, on multivariate regression, only pediatric versus adult patient remained predictive of events ($P = .02$; rate of 21.3% for pediatric patients vs 10.8% for adult patients). For the other 3 treatment techniques, time from simulation to first treatment was not significantly associated with near-miss events, nor were other clinical factors. Figure 1 illustrates the relationship between planning time and near-miss events in the four treatment type groups.

The large sample size of the 3D conformal and IMRT groups permitted a more detailed regression analysis incorporating time in each planning step instead of overall time from simulation to first treatment (Table 5). On multivariate analysis, no planning step was associated with near-miss events.

Delayed treatments; patient safety events

The preceding analysis was based on the actual time between completion of planning steps, not the originally planned time. Treatments whose start dates need to be delayed because of issues with treatment planning could have higher risk of near-miss events. At our institution, at the time of simulation, the patient is scheduled for his or her verification simulation return visit. When treatment needs to be delayed because of planning or clinical issues, the verification simulation appointment is canceled and a new, later appointment is created. Delayed treatments and the length of treatment delay were recorded by searching the OIS for treatments with a canceled verification simulation appointment. Of the 2257 treatment courses analyzed, 212 (9.4%) were delayed from planned start by a mean of 4.2 days (range, 1-34 days). The originally planned time from simulation to verification simulation was longer in treatments that ended up being delayed, with a mean of 7.3 versus 5.9 days. The longest delays were generally related to patient factors such as acute illness. Treatment delay did not have a significant association with serious near-miss events: 15.1% of delayed treatments had a grade 3-4 near-miss event, as opposed to 11.5% of nondelayed treatments ($P = .15$ by χ^2 test).

It would be useful to understand factors leading to serious errors that reached the patient. However, only 8 of

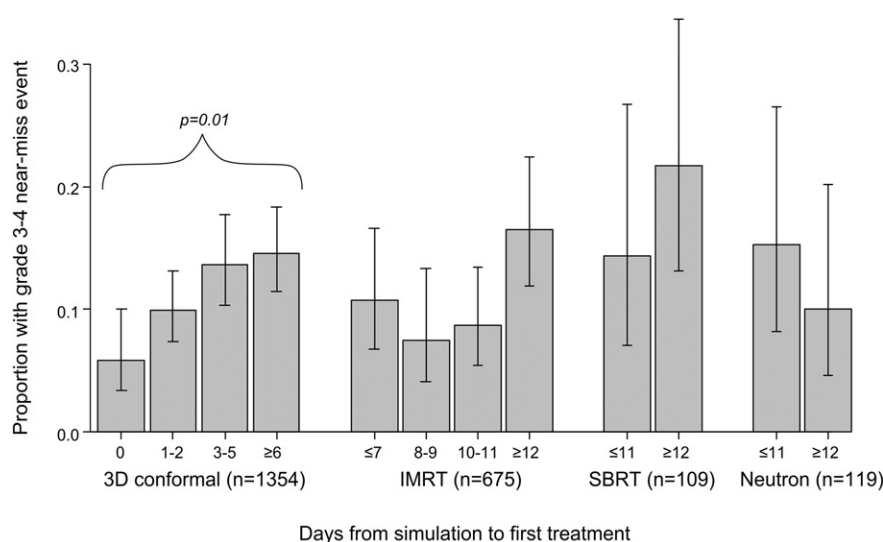


Figure 1 Risk of severe near-miss events by treatment technique and days from simulation to first treatment. Time from simulation to treatment affected event rate only in the 3-dimensional (3D) conformal group ($P = .01$). 3D conformal and intensity modulated radiation therapy (IMRT) groups divided into time quartiles; stereotactic body radiation therapy (SBRT) and neutron groups are divided into greater than/less than median time because of those groups' smaller sample size. Error bars indicate 95% confidence interval.

the 2257 treatment courses in this study (0.4%) had an error that reached the patient, all of which were no-harm events. With so few events, we could not perform a separate statistical analysis to identify factors predictive of serious errors that reached the patient.

Discussion

By studying near-miss events that occur during radiation treatment planning and delivery, high-risk procedures and clinical situations can be targeted for process improvements, with the goal of preventing serious

errors and patient harm.⁹ A recent study indicated that treatments with a short time allotted to treatment planning had more errors; in another report, the most common contributing factor for incidents was haste.^{4,5} Although these studies suggest that compressed treatment planning schedules are detrimental to patient safety, a detailed risk analysis of planning time that takes into account treatment technique/complexity has not been reported.

By linking our department's incident learning system and oncology information system, we generated a large dataset to explore this topic. In separate multivariate regressions for the 4 treatment techniques studied, time spent in treatment planning was not a significant predictor

Table 5 Predictors of treatments with grade 3-4 near-miss events, including time spent in each phase of the planning process

Treatment type	Characteristic	Univariate		Multivariate	
		OR (95% CI)	P value	OR (95% CI)	P value
3D conformal (n = 1354)	Days from simulation to contour completion	1.07 (1.00-1.13)	.04	1.01 (0.94-1.08)	.69
	Days from contour completion to plan completion	1.08 (1.01-1.15)	.01	1.03 (0.96-1.10)	.39
	Days from plan completion to virtual simulation	1.01 (0.94-1.07)	.76	0.99 (0.91-1.06)	.78
	Days from virtual simulation to first treatment	1.03 (0.95-1.10)	.35	1.02 (0.93-1.09)	.50
	Metastasis treated	0.52 (0.36-0.74)	<.001	0.71 (0.47-1.09)	.12
	Treatment fractions	1.03 (1.01-1.04)	<.001	1.02 (1.00-1.03)	.09
	Pediatric patient	2.25 (1.33-3.66)	.002	1.84 (1.07-3.05)	.02
IMRT (n = 675)	Days from simulation to contour completion	0.99 (0.88-1.09)	.83	1.01 (0.90-1.11)	.91
	Days from contour completion to plan completion	1.04 (0.94-1.14)	.42	1.04 (0.94-1.14)	.42
	Days from plan completion to virtual simulation	1.07 (0.98-1.16)	.08	1.07 (0.98-1.16)	.09
	Days from virtual simulation to first treatment	1.16 (0.93-1.46)	.16	1.16 (0.93-1.49)	.17
	Metastasis treated	1.19 (0.40-2.89)	.73	1.21 (0.37-3.31)	.73
	Treatment fractions	1.00 (0.98-1.02)	.98	1.00 (0.98-1.02)	.86
	Pediatric patient	0.54 (0.09-1.84)	.41	0.63 (0.10-2.24)	.54

SBRT and neutron subsets were not included due to insufficient sample size. Bolded P value indicates $P < .05$.

3D, 3-dimensional; CI, confidence interval; IMRT, intensity modulated radiation therapy; OR, odds ratio; SBRT, stereotactic body radiation therapy.

of severe near-miss events. Instead, near-miss event probability varied by treatment type, with the highest rate occurring in the most complex treatment type of SBRT (18% vs 11%). Within the largest of the 4 technique groups, 3D conformal, treatment of pediatric patients was riskier, with event rate double that of adult patients. These results point toward treatment complexity as a culprit in radiation therapy errors: clinical complexity in the case of pediatric patients and technical complexity in the case of SBRT. This finding is consistent with other studies, which also found a higher error rate for more complex treatments and also specifically for pediatric cases.^{4,5}

Why was there no significant relationship between planning time and near-miss events in the present study when a prior study⁵ showed more errors for rapidly planned treatments? There may be process differences between departments that affect error rates in various situations. For instance, in 1 department, there could be more standardization of isocenter/skin marks for rapidly planned palliative treatments. Another difference with prior studies is that our analysis explicitly took the interaction between planning time and treatment type into account. Finally, in the Walker et al⁵ report, 1.4% of treatments had a reported incident, whereas in this study 11.8% of treatments had a grade 3-4 incident. This suggests that a different spectrum of error severity was included, which could affect results. It is important to note that a higher rate of near-miss incident reporting does not imply an unsafe environment; in fact, it has been linked to lower rate of treatment complications, likely because it indicates presence of a healthy safety culture.¹³

Does the present analysis suggest any actionable targets for safety improvements? Our results do not give strong evidence for interventions such as “no-fly” process, in which a delay in 1 planning step automatically leads to the treatment start date being proactively pushed back.^{3,8} On the other hand, the large variation in near-miss event rates observed between treatment types and patient age groups highlights the value of an incident learning system in identifying trends affecting patient safety.

This study has some limitations. First, from the data recorded in our department’s OIS, it is possible to determine when each treatment planning task was completed, but not the number of hours of work spent on that task or how rushed that work was. Second, a compressed planning time could result in less chance to enter incident reports because staff members would not have time to step away from their work and enter a report. It is unlikely that this explains the lack of increased incidents seen with rapidly planned treatments, because the patterns were similar when either all events or only severe events were examined (data not shown). If selective reporting were occurring, one would expect that severe events would be more likely to be reported even when staff

members are rushed. Finally, the current study does not specifically address emergency treatments, such as those occurring on nights and weekends with planning time measured in hours instead of days. We previously reported that these emergency treatments have a higher ratio of severe to minor near-miss events compared with none-emergent treatments.¹⁴ Therefore, in the extreme, rushed work clearly has the potential to induce errors, as has been found in other medical fields.^{6,7}

In conclusion, treatments with compressed planning schedules did not have an increased risk of near-miss events when treatment complexity was taken into account. These results suggest that staff members are able to maintain high quality work even when under moderate time pressure. The technique and complexity of treatments should be taken into account when targeting safety interventions.

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